AI Code Quality Auto-Evaluation: A Diff-Based Agentic Workflow for SOLID Principle Violation Detection

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Introduction & Motivation

Modern software increasingly relies on LLMs for code generation. Ensuring adherence to **SOLID design principles** is critical but **manual evaluation does not scale** to production codebases. We automate SOLID violation detection using a diff-based agentic workflow.

SOLID Principles

- **S**ingle Responsibility — one reason to change
- **O**pen / Closed — open for extension, closed for modification
- **L**iskov Substitution — subtypes substitutable for base types
- **I**nterface Segregation — prefer many specific interfaces
- **D**ependency Inversion — depend on abstractions

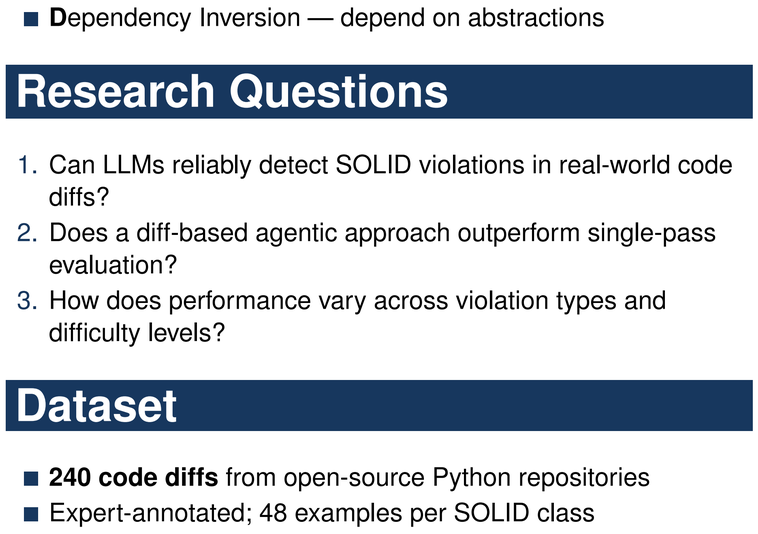
Research Questions

1. Can LLMs reliably detect SOLID violations in real-world code diffs?
2. Does a diff-based agentic approach outperform single-pass evaluation?
3. How does performance vary across violation types and difficulty levels?

Dataset

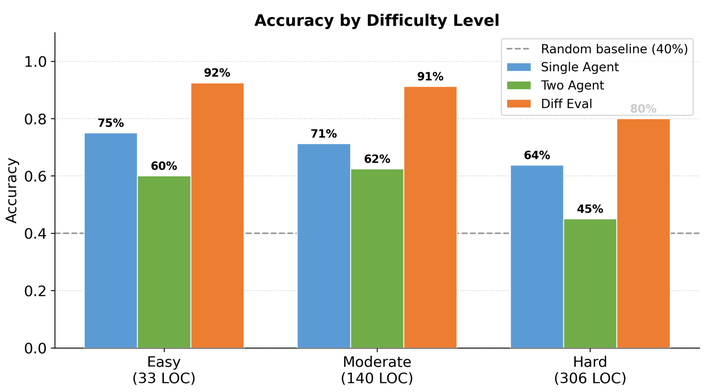
- **240 code diffs** from open-source Python repositories
- Expert-annotated; 48 examples per SOLID class
- Difficulty: Easy (33 LOC), Moderate (140 LOC), Hard (306 LOC)

Results: Overall Accuracy



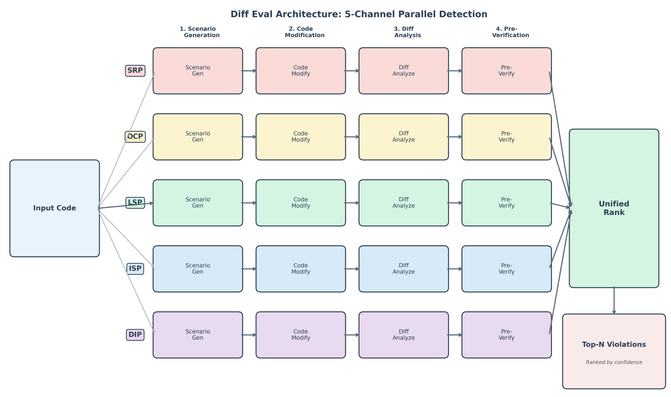
Diff Eval achieves **87.9%**, outperforming Single Agent (70.0%) and Two Agent (55.8%) by **+17.9 pp**.

Results: Accuracy by Difficulty



Diff Eval maintains $\geq$80% even on hard examples (306 LOC).

System Architecture



Five parallel sub-agents each specialise in one SOLID principle. Each channel independently generates scenarios, modifies code, analyzes the resulting diff, and pre-verifies its verdict before a unified ranking resolves conflicts.

Diff-Based Evaluation Pipeline

1. **Scenario Generation** — Enumerate plausible violation scenarios for the target principle.
2. **Code Modification** — Minimally transform code to embody the violation
3. **Diff Analysis** — Confirm structural signals in the resulting diff
4. **Pre-Verification** — Self-consistency check before unified ranking

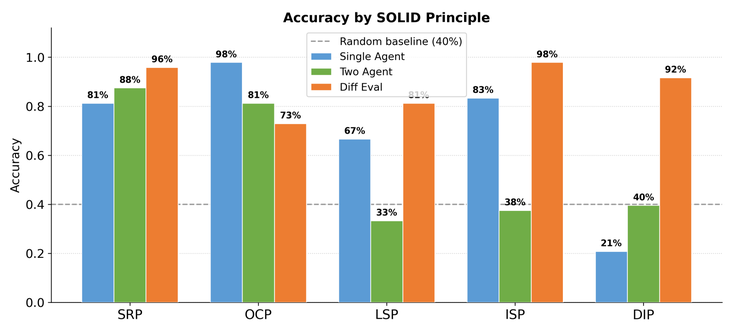
Why Diff-Based?

- Focuses on *changes* introduced by a commit, not pre-existing style
- Reduces false positives; mirrors pull-request code review
- Provides fine-grained, explainable output per principle

Baseline Comparisons

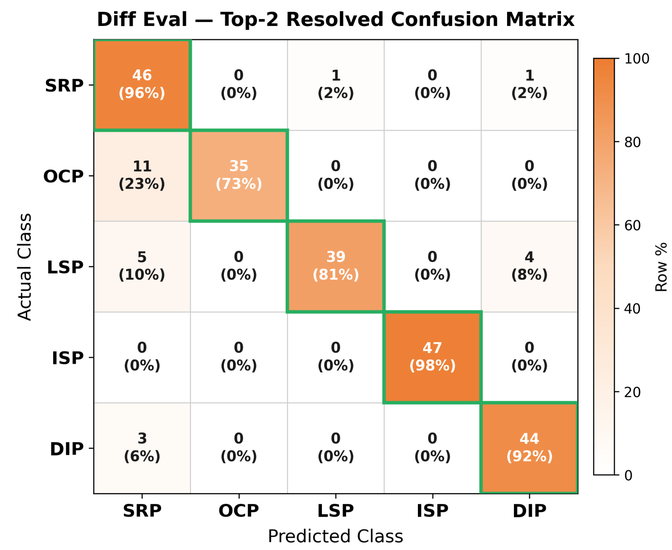
Method	Agents	Diff	Accuracy
Single Agent	1	×	70.0%
Two Agent	2	×	55.8%
Diff Eval	**5**	✓	**87.9%**
Random Baseline	—	—	40.0%

Accuracy by SOLID Principle



ISP (98%) and SRP (96%) reach near-perfect accuracy. LSP (81%) remains the most challenging, reflecting its subtle type-substitution semantics.

Confusion Matrix — Diff Eval



Note: 4 examples with no valid prediction excluded.

Strong diagonal dominance confirms the model reliably distinguishes all five SOLID classes. OCP-to-SRP confusion (23%) is the main error mode, reflecting structural overlap between the two principles.

Key Findings

- Diff-based agentic workflow achieves **87.9% accuracy**, far above the 40% random baseline
- Specialised per-principle agents are critical: the generalist Two Agent baseline drops **14 pp** vs. Single Agent
- Performance is **robust to code length** — hard examples (306 LOC) still reach 80%
- ISP and SRP are easiest; **LSP remains the hardest** class across all methods

Conclusion

We demonstrate that a diff-based multi-agent LLM pipeline can **automate SOLID violation detection at scale**, matching expert-level precision on a diverse benchmark.

The diff-based framing is the decisive design choice: by constraining analysis to code changes, the system focuses on violations *introduced* by a commit, eliminating noise and yielding highly interpretable, actionable reports.

Future Work

- **CI/CD integration** — real-time SOLID checks on pull requests
- Multi-language support beyond Python (Java, TypeScript)
- Extension to broader design pattern detection
- Fine-tuning smaller models on the annotated benchmark

Acknowledgements

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